Question Paper

Physics (Maximum Marks: 90)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A lead bullet penetrates into a solid object and melts. Assuming that 50% of its kinetic energy was used to heat it, the initial speed of the bullet is (the initial temperature of the bullet is 25^oC and its melting point is 300^oC). Latent heat of fusion of lead $2.5 \times 10^4\ J/kg$ and specific heat capacity of lead $= 125\ J/kg - K$

A. $100\ m/s$
B. $\approx 490 m/s$
C. $520\ m/s$
D. $360\ m/s$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The dimensional formula for angular momentum is:

A. $[ML^2T^{-2}]$

B. $[MLT^{-1}]$

C. $[ML^2T^{-1}]$

D. $[M^0L^2T^{-2}]$

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In Young's double slit experiment, one of the slit is wider than other, so that the amplitude of the light from one slit is double of that from other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by

A. $\frac{I_m}{3}(1 + 2\cos^2\frac{\phi}{2})$

B. $\frac{I_m}{5}(1+4\cos^2\frac{\phi}{2})$

C. $\frac{I_m}{9}(1+8\cos^2\frac{\phi}{2})$

D. $\frac{I_m}{9}(4+5\cos\phi)$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The gravitational acceleration on the surface of the earth of radius R and mean density ρ is

A. $\frac{4}{3}G\pi R^2\rho$

B. $\frac{4}{3}G\pi^2R^2\rho$

C. $\frac{2}{3}G\pi R\rho$

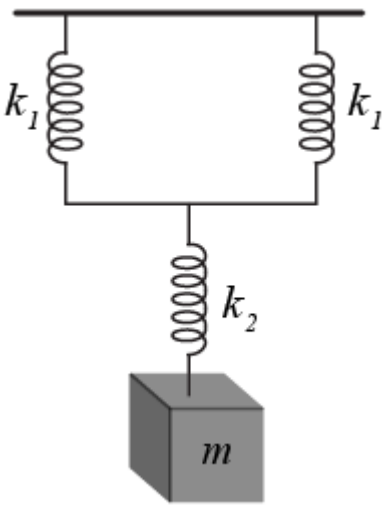
D. $\frac{4}{3}G\pi R\rho$

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What will be the force constant of the spring system shown in the figure?



A. $\frac{k_1}{2}+k_2$

B. $[\frac{1}{2k_1}+\frac{1}{k_2}]^{-1}$

C. $\frac{1}{2k_1}+\frac{1}{k_2}$

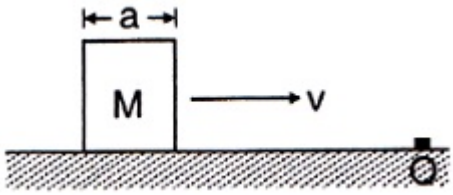
D. $[\frac{2}{k_1}+\frac{1}{k_2}]^{-1}$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A cubical bock of side α moving with velocity v on a horizontal smooth plane as shown. It hits a ridge at point O. The angular speed of the block after it hits O is



A. $3v / 4a$
B. $3v / 2a$
C. $\sqrt{\frac{3}{2}} \frac{v}{a}$
D. Zero

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Length of 20 cm (exact) long pipe is measured by two instruments and reported as $19.65\ cm$ and $20.1\ cm$

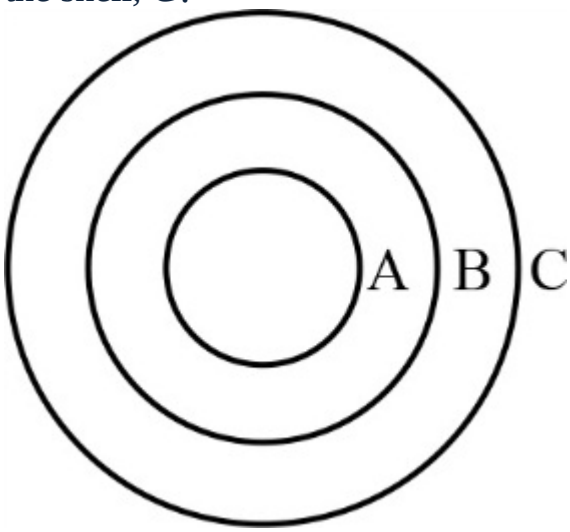
A. 19.65 cm is more accurate
B. Both measurements are equally precise
C. Both measurements are equally accurate
D. 20.1 cm is less precise

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three concentric spherical shells, $A,\ B\ \&\ C$ have charges $Q,\ 2Q, -Q$ respectively. Calculate the ratio of charges on the inner and the outer surfaces of the shell, C.



A. $+\frac{3}{4}$
B. $-\frac{3}{4}$
C. $\frac{3}{2}$
D. $-\frac{3}{2}$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The self inductance of a coil is L. Keeping the length and area same, the number of turns in the coil is increased to four times. The self inductance of the coil will now be

A. $\frac{1}{4} L$
B. L
C. $4L$
D. $16L$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two bar magnets having same geometry with magnetic moments m and $2m$, are firstly placed in such a way that their similar poles are on same side. Then, the time period of oscillation is T_1. Now, the polarity of one of the magnet is reversed. Then, the time period of oscillation is T_2, this implies

A. $T_1 < T_2$
B. $T_1 = T_2$
C. $T_1 > T_2$
D. $T_2 = \infty$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A large parallel plate capacitor, whose plates have an area of $1\ \mathrm{m}^2$ and are separated from each other by $1\ \mathrm{mm}$, is being charged at a rate of $25\ \mathrm{V\ s}^{-1}$. If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is

A. $25\ \mu\mathrm{A}$
B. $11\ \mu\mathrm{A}$
C. $2.2\ \mu\mathrm{A}$
D. $1.1\ \mu\mathrm{A}$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If lower half of the body of a lens is covered with black paper, the image produced by the lens will

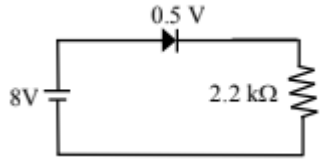
A. disappear.
B. remain almost unaffected.
C. have its size reduced to half.
D. have its intensity reduced to half.

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the circuit shown, if the forward voltage drop for the diode is $0.5\ \mathrm{V}$, then the approximate current in the circuit will be



A. $3.4\ \mathrm{mA}$
B. $2\ \mathrm{mA}$
C. $2.5\ \mathrm{mA}$
D. $3\ \mathrm{mA}$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The power factor of a RC circuit is $\frac{1}{\sqrt{2}}$. If the frequency of AC signal is halved, then the power factor of the circuit becomes:

A. $\frac{1}{\sqrt{5}}$
B. $\frac{1}{\sqrt{7}}$
C. $\frac{1}{\sqrt{3}}$
D. $\frac{1}{\sqrt{11}}$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Critical velocity is given by (N = Reynold's number)

A. $v_C = \frac{N\eta}{\rho D}$
B. $v_C = \frac{N\eta\rho}{D}$
C. $v_C = \frac{N\eta D}{\rho}$
D. $v_C = \frac{N}{\eta\rho D}$

Question No. 16
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have 12 μC and $-3\ \mu C$ charges, respectively, and are at a large distance from each other, They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively:

A. $4.5\ \mu C$ of both
B. $+4.5\ \mu C$ and $-4.5\ \mu C$
C. $3\ \mu C$ and $6\ \mu C$
D. $6\ \mu C$ and $3\ \mu C$

Question No. 17
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

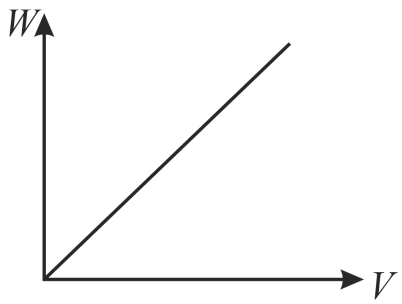
A body is projected with an initial velocity of $(8\hat{i} + 6\hat{j})$ m s^{-1} . The horizontal range is (g = 10 m s^{-2})

A. 9.6 m
B. 14 m
C. 50 m
D. 100 m

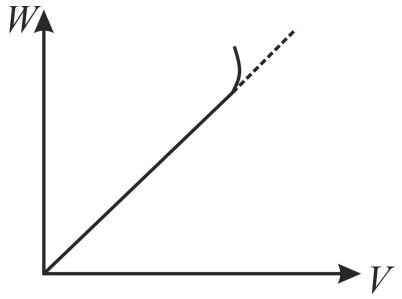
Question No. 18
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

A particle, initially at rest on a frictionless horizontal surface, is acted upon by a horizontal force which is constant in size and direction. A graph is plotted between the work done (W) on the particle, against the speed of the particle (v). Let there be no other horizontal forces acting on the particle. What will the graph possibly be looking like ?

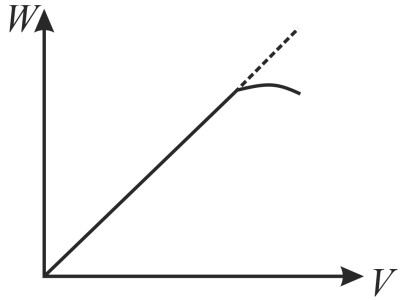
A.



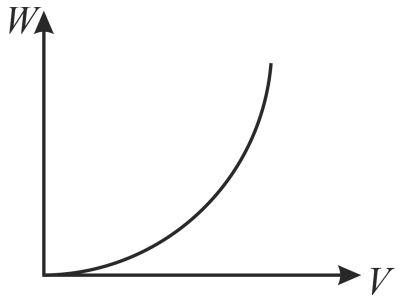
B.



C.



D.

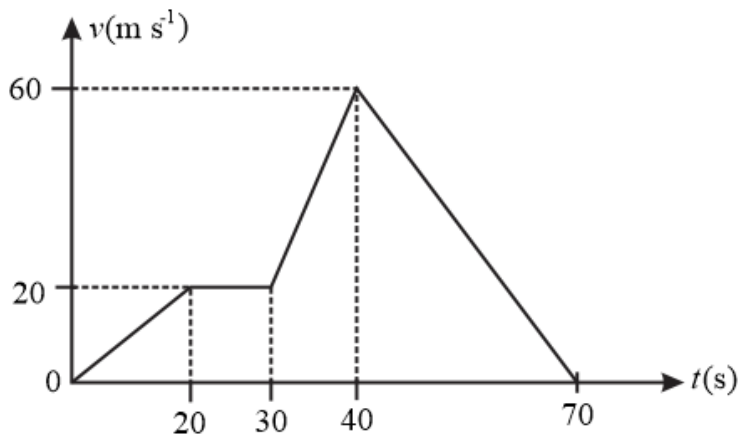


Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The $v-t$ graph of a body is given in figure. The maximum acceleration in $\mathrm{m\ s^{-2}}$ is



A. 4
B. 3
C. 2
D. 1

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The inductance of a closed-packed coil of 400 turns is $8\ \mathrm{mH}$. A current of $5\ \mathrm{mA}$ is passed through it. The magnetic flux through each turn of the coil is

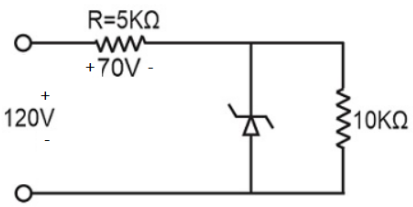
A. $\frac{1}{4\pi}\mu_0\mathrm{Wb}$
B. $\frac{1}{2\pi}\mu_0\mathrm{Wb}$
C. $\frac{1}{3\pi}\mu_0\mathrm{Wb}$
D. $0.4\mu_0\mathrm{Wb}$

Question No. 21

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the circuit shown in the figure :



A. Current through Zener diode is 4 mA
B. Current through Zener diode is 9 mA
C. The output voltage is 60 V
D. The output voltage is 40 V

Question No. 22

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle executing SHM crosses points A and B with the same velocity. Having taken $3s$ in passing from A to B, it returns to B after another $3s$. The time period is

A. $15s$
B. $6s$
C. $12s$
D. $9s$

Question No. 23

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

70 cal of heat is required to raise the temperature of 2 moles of an ideal gas from 30 $^\circ C$ to 35 $^\circ C$ while the pressure of the gas is kept constant. The amount of the heat required to raise the temperature of the same gas through the same temperature range at constant volume is (gas constant $R = 2 \text{ cal mol}^{-1} - \text{K}^{-1}$)

A. 70 cal
B. 60 cal
C. 50 cal
D. 30 cal

Question No. 24

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following has the mass closest in value to that of the positron?

A. Proton
B. Electron
C. Photon
D. Neutrino

Question No. 25

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The angle turned by a body undergoing circular motion depends on time as, $\theta = \theta_0 + \theta_1 t + \theta_2 t^2$, where θ_0, θ_1, θ_2 are constants and t = time . Then, the angular acceleration of the body is,

A. θ_1
B. θ_2
C. $2\theta_1$
D. $2\theta_2$

Question No. 26

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The half-life of ^{131}I is 8 days. Given a sample of ^{131}I at time $t = 0$, we can assert that

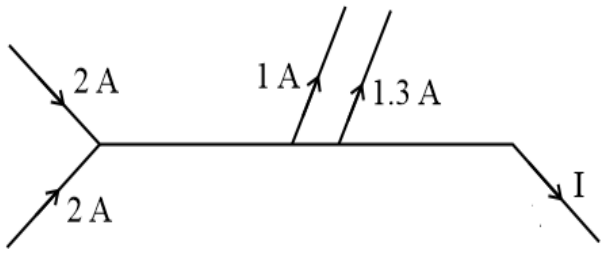
A. No nucleus will decay before $t = 4$ days
B. No nucleus will decay before $t = 8$ days
C. No nucleus will decay before $t = 16$ days
D. A given nucleus may decay at any time after $t = 0$

Question No. 27

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of current I in the circuit will be



A. 1.7 A
B. 2.1 A
C. 3 A
D. 0

Question No. 28

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The energy of incident photons corresponding to the maximum wavelength of visible light is:

A. 3.2 eV
B. 7 eV
C. 1.6 eV
D. 1 eV

Question No. 29

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The radii of two planets are respectively R_1 & R_2 and their densities are respectively ρ_1 & ρ_2. The ratio of the acceleration due to gravity at their surface is -

A. $g_1 : g_2 = \frac{\rho_1}{R_1^2} : \frac{\rho_2}{R_2^2}$
B. $g_1 : g_2 = R_1R_2 : \rho_1\rho_2$
C. $g_1 : g_2 = R_1\rho_2 : R_2\rho_1$
D. $g_1 : g_2 = R_1\rho_1 : R_2\rho_2$

Question No. 30

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A metallic rod breaks when strain produced is 0.2%. Young's modulus of the material of the rod is 7×10^9 N m^{-2} . What should be its area of cross-section to support a load of 10^4 N?

A. 7.1×10^{-8} m^2
B. 7.1×10^{-6} m^2
C. 7.1×10^{-4} m^2
D. 7.1×10^{-2} m^2

Chemistry (Maximum Marks: 90)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

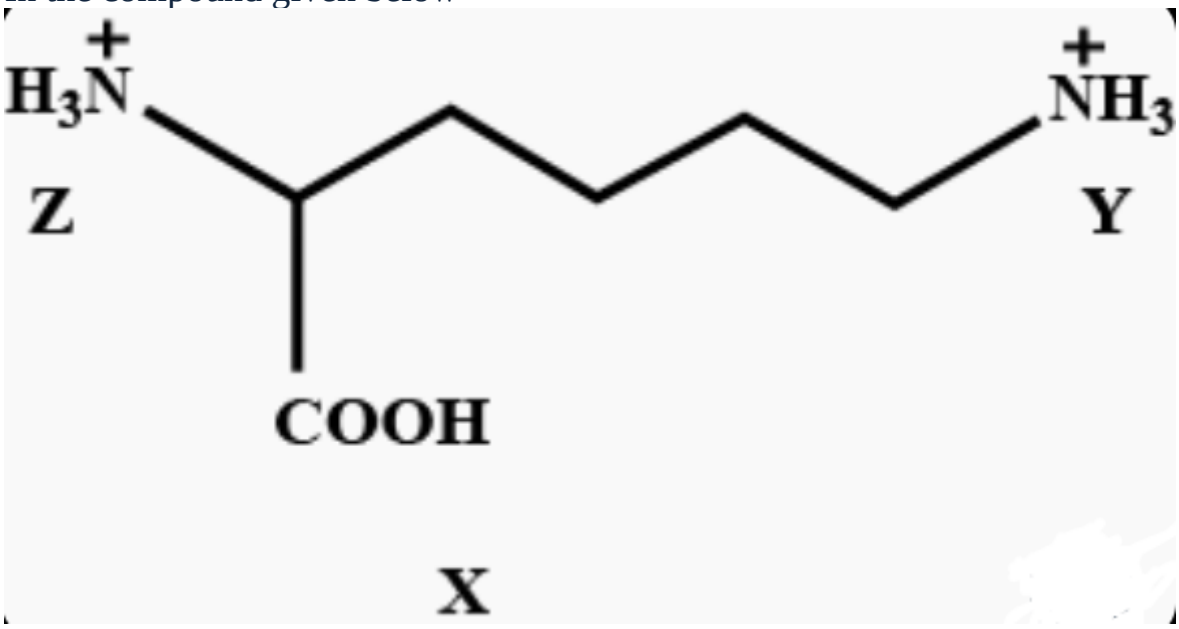
The reaction, $x \rightarrow$ product follows first order kinetics. In 40 minutes the concentration of X changes from $0.1M$ to $0.025M$ Then the rate of reaction when concentration of X is 0.01M

A. 1.73×10^{-4} Mmin^{-1}
B. 3.47×10^{-5} Mmin^{-1}

C. 3.47×10^{-4} Mmin^{-1}
D. 1.73×10^{-5} Mmin^{-1}

Question No. 2
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

In the compound given below



The correct order of the acidity of the positions $(X), (Y)$ and (Z) is

A. $(Z) > (X) > (Y)$
B. $(X) > (Y) > (Z)$
C. $(X) > (Z) > (Y)$
D. $(Y) > (X) > (Z)$

Question No. 3
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are:

A. $4.90,\ 0$ and 1.73 BM
B. $4.90,\ 0$ and 2.83 BM
C. $5.82,\ 0$ and 0 BM
D. $5.92,\ 4.90$ and 0 BM

Question No. 4
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The number of hydroxyl group in pyrophosphoric acid is

A. 3
B. 4
C. 5
D. 7

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What is Z in the following sequence of reactions?

$$\text{Phenol} \xrightarrow[\text{Dust}]{\text{Zn}} X \xrightarrow[\text{Anhydrous } AlCl_3]{CH_3Cl\ /} Y \xrightarrow[KMnO_4]{\text{Alkaline}} Z$$

A. Benzene
B. Toluene
C. Benzaldehyde
D. Benzoic acid

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The reaction

$CH_3COOH + Cl_2 \xrightarrow{P} ClCH_2COOH + HCl$ is called

A. Hell-Volhard-Zelinsky reaction
B. Birch reaction
C. Rosenmund reaction
D. Hunsdiecker reaction

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following alkyl groups has the maximum $+I$ effect

A. CH_3-
B. $(CH_3)_2CH-$
C. $(CH_3)_3C-$
D. CH_3CH_2-

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The common oxidation state of the elements of lanthanide series is

A. $+1$
B. $+3$
C. $+4$
D. $+6$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct statement regarding defects in solids is-

A. Frenkel defect is favoured by a very small difference in the size of cation and anion.
B. Frenkel defect is not a dislocation defect.
C. Trapping of e^- in lattice leads to the formation of F-center.
D. Schottky defects have no effect on the physical properties of solid.

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

On adding 0.750gm of compound in 25gm of solvent lowered the freezing point of the solvent by 0.502°C. The molecular wt. of the substance is [the molecular depression constant = 50.2°C per 100gm of solvent]

A. 100
B. 200
C. 300
D. 400

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For an exothermic reaction

A. H of the products is less than H of the reactants
B. H of the products is more than H of the reactants
C. H of the products is equal to H of the reactants
D. ΔH is always positive

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The series limit for Balmer series of H-spectrum is

A. 3800
B. 4200
C. 3646
D. 4000

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Three alcohols
(i) $CH_3CH_2CH_2OH$,
(ii) $CH_3 - CHOH - CH_3$ and
(iii) $CH_3 - C(CH_3)(OH) - CH_3$
were treated seperately with Lucas reagent (Conc. HCl + $ZnCl_2$). What results do you expect at room temperature

A. (i) reacts in about 5 minutes, (ii) reacts in about 15 minutes and (iii) not at all
B. (iii) reacts immediately, (ii) reacts in about 5 minutes and (i) not at all
C. (ii) and (iii) react immediately and (i) in about 5 minutes
D. (i) reacts immediately, (ii) reacts in about 5 minutes and (iii) not at all

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Physical adsorption

A. Involves the weak attractive interaction between the adsorbent and adsorbate
B. Involves the chemical interactions between the adsorbent and adsorbate
C. Is irreversible in nature
D. Increases with increase of temperature

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

When N_2 goes to N_2^+, the $N - N$ bond distance ..... and when O_2 goes to O_2^+, the $O - O$ bond distance .....

A. Decrease, increase
B. Increase, decrease
C. Increase, increase
D. None of these

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The formula of dichloro bis (urea) copper (II) is

A. $[Cu\{O = C(NH_2)_2\}Cl_2]$
B. $[CuCl_2\{O = C(NH_2)_2\}_2]$
C. $[Cu\{O = C(NH_2)_2\}Cl]Cl$
D. $[CuCl_2]\{O = C(NH_2)_2H_2\}$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Ionisation energy in group I-A varies in the decreasing order as

A. $Li > Na > K > CS$
B. $Na > Li > K > CS$
C. $Li > CS > K > Na$
D. $K > CS > Na > Li$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One mole of an ideal gas ($C_V = 20JK^{-1} mol^{-1}$) initially at STP is heated at constant volume to twice the initial temperature. For the process, W and q will be-

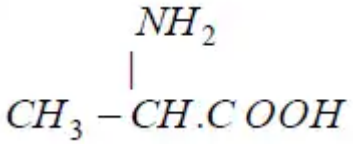
A. $W = 0; q = 5.46kJ$
B. $W = 0; q = 0$
C. $W = -5.46kJ; q = 5.46kJ$
D. $W = 5.46kJ; q = 5.46kJ$

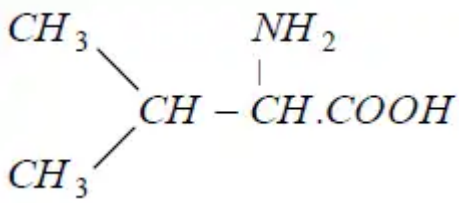
Question No. 19

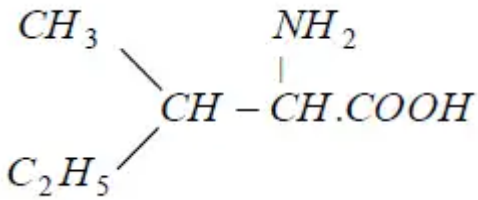
Only One Option Correct Type

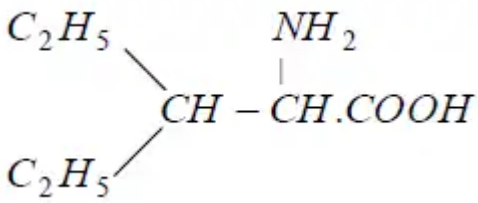
Each question has multiple options out of which ONLY ONE is correct.

The Structural formula of an amino acid, isoleucine is

A. 

B. 

C. 

D. 

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following compounds shows optical isomerism

A. $CH_3CHCl - CH_2 - CH_3$
B. $CH_3 - CH_2 - CHCl - CH_2 - CH_3$
C. $ClCH_2 - CH_2 - CH_2 - CH_3$
D. $ClCH_2 - CH_2 - CH_3$

Question No. 21

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following ions gives coloured solution

A. Cu^+
B. Zn^{++}
C. Ag^+

D. Fe^{2+}

Question No. 22

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The role of conc. H_2SO_4 in the esterification process is

A. Catalyst
B. Dehydrating agent
C. Hydrolysing agent
D. Dehydrating agent and catalyst

Question No. 23

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Aspirin is

A. Antibiotic
B. Antipyretic
C. Sedative
D. Psychedelic

Question No. 24

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Time taken for an electron to complete one revolution in the Bohr orbit of hydrogen atom is

A. $\frac{4\pi^2 mr^2}{nh}$
B. $\frac{nh}{4\pi^2 mr}$
C. $\frac{nh}{4\pi^2 mr^2}$
D. $\frac{h}{2\pi mr}$

Question No. 25

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which are the monomers, used in the production of nylon − 6, 6?

A. Hexamethylene diamine and ethylene glycol
B. Adipic acid and ethylene glycol
C. Adipic acid and hexamethylene diamine

D. Dimethyl terephthalate and ethylene glycol

Question No. 26

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Laughing gas is prepared by heating

A. NH_4Cl
B. $(NH_4)_2SO_4$
C. $NH_4Cl + NaNO_3$
D. NH_4NO_3

Question No. 27

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The standard EMF for the given cell reaction $Zn + Cu^{2+} \Rightarrow Cu + Zn^{2+}$ is 1.10 V at 25°C. The EMF for the cell reaction, when $0.1M Cu^{2+}$ and $0.1M\ Zn^{2+}$ solutions are used, at 25°C is

A. 1.10 V
B. 0.110 V
C. −1.10 V
D. −0.110 V

Question No. 28

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For which one of the following reactions $K_p = K_c$

A. $N_2 + 3H_2 \rightleftharpoons 2NH_3$
B. $N_2 + O_2 \rightleftharpoons 2NO$
C. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$
D. $2SO_3 \rightleftharpoons 2SO_2 + O_2$

Question No. 29

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For reactions $A \rightarrow B$ and $P \rightarrow Q$ Arrhenius constants are 10^8 and 10^{10} respectively. If $E_{A \rightarrow B} = 600$ cal/mole and $E_{P \rightarrow Q} = 1200$ cal/mole, then find the temperature at which their rate constants are same.
(Given: $R = 2$ cal/mole/K)

A. 600 K
B. 300×4.606 K
C. $\frac{300}{4.606}$ K
D. $\frac{4.606}{600}$ K

Question No. 30
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

Which of the following alkali metals is smallest in size

A. Rb
B. k
C. Na
D. Li

Mathematics (Maximum Marks: 120)

Question No. 1
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The function $f : R \rightarrow R$ defined as $f(x) = \frac{x^2-x+1}{x^2+x+1}$ is

A. injective as well as surjective
B. injective but not surjective
C. surjective but not injective
D. neither injective nor surjective

Question No. 2
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

If $x, y \in [0,2\pi]$, then total number of ordered pairs (x, y) satisfying the equation $\sin x . \cos y = 1$, is equal to

A. 1
B. 2
C. 3
D. 7

Question No. 3
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The maximum value of $z = 3x + 2y$ subject to
$x + 2y \geq 2, x + 2y \leq 8, x, y \geq 0$ is :

A. 32
B. 24
C. 40
D. None of these

Question No. 4
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The distance of the point $(1, -2, 4)$ from the plane passing through the point $(1, 2, 2)$ and perpendicular to the planes $x - y + 2z = 3$ and $2x - 2y + z + 12 = 0$, is :

A. 2
B. $\sqrt{2}$
C. $2\sqrt{2}$
D. $\frac{1}{\sqrt{2}}$

Question No. 5
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

If the area of the triangle on the complex plane formed by the 3 points $z, z + iz$ and iz is 200, then the value of $3|z|$ must be equal to

A. 20
B. 40
C. 60
D. 80

Question No. 6
Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

There is a set of m parallel lines intersecting a set of another n parallel lines in a plane. The number of parallelograms formed, is

A. ${}^{m-1}C_2 . {}^{n-1}C_2$
B. ${}^{m}C_2 . {}^{n}C_2$
C. ${}^{m-1}C_2 . {}^{n}C_2$
D. ${}^{m}C_2 . {}^{n-1}C_2$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let a, b and c be the 7th, 11th and 13th terms respectively of a non-constant A.P. . If these are also the three consecutive terms of a G.P. , then $\frac{a}{c}$ is equal to:

A. 2
B. $\frac{7}{13}$
C. $\frac{1}{2}$
D. 4

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A straight line L through the point (3, -2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is

A. $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$
B. $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
C. $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$
D. $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is

A. 0
B. 1
C. 3
D. 4

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

One of the directrices of the ellipse $8x^2 + 6y^2 - 16x + 12y + 13 = 0$ is

A. $3y - 3 = \sqrt{6}$
B. $3y + 3 = \sqrt{6}$
C. $y + 1 = \sqrt{3}$

D. $y-1=-\sqrt{3}$

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\lim_{x \to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{\frac{2}{1+\sin x-\cos x}}$ is equal to (where $a>0,\ b>0,\ c>0$)

A. (abc)
B. $(abc)^{\frac{3}{2}}$
C. $(abc)^{\frac{2}{3}}$
D. None of these

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The mean and the median of the following ten numbers in increasing order $10,\ 22,\ 26,\ 29,\ 34,\ x,\ 42,\ 67,\ 70,\ y$ are 42 and 35 respectively, then $\frac{y}{x}$ is equal to:

A. $\frac{9}{4}$
B. $\frac{7}{3}$
C. $\frac{7}{2}$
D. $\frac{8}{3}$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If m and σ^2 are the mean and variance of the random variable X . whose distribution is given by

x	0	1	2	3
$P(x)$	1/3	1/2	0	1/6

then

A. $m=\sigma^2=2$
B. $m=1, \sigma^2=2$
C. $m=\sigma^2=1$
D. $m=2, \sigma^2=1$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The greatest integer which divides the number $101^{100} - 1$ is

A. 100
B. 1000
C. 10000
D. 100000

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The curve represented by $4x^2 - 4y^2 - 6xy + 5x + 5y = 7$ is:

A. A parabola
B. A hyperbola
C. A pair of straight lines
D. An ellipse

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x) = x^n + 4$, then the value of $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \ldots + \frac{f^n(1)}{n!}$ is

A. 2^{n-1}
B. $2^n + 4$
C. $1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{2!} + \ldots + \frac{1}{n!}$
D. None of these

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Number of solution(s) of the equation $\sin^{-1}|x| = |\cos^{-1} x|$, is/are

A. 0
B. 1
C. 2
D. 3

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $AB =$

A. B
B. $5B$
C. B^5
D. $5A$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider the family of lines $(x + y - 1) + \lambda(2x + 3y - 5) = 0$ and $(3x + 2y - 4) + \mu(x + 2y - 6) = 0$, equation of a straight line that belongs to both the families is

A. $x - 2y - 8 = 0$
B. $x - 2y + 8 = 0$
C. $2x + y - 8 = 0$
D. $2x - y - 8 = 0$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let α and β be the roots of equation $px^2 + qx + r = 0, p \neq 0$. If p, q and r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is

A. $\frac{\sqrt{61}}{9}$
B. $\frac{2\sqrt{71}}{9}$
C. $\frac{\sqrt{34}}{9}$
D. $\frac{2\sqrt{13}}{9}$

Question No. 21

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $x\frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the differential equation is

A. $\log \frac{x}{y} = Cy$
B. $\log \frac{y}{x} = Cy$
C. $\log \frac{x}{y} = Cx$
D. $\log \frac{y}{x} = Cx$

Question No. 22

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\vec{p}$, $\vec{q}$ are two non-collinear and non-zero vectors such that $(b-c)\vec{p} \times \vec{q} + (c-a)\vec{p} + (a-b)\vec{q} = 0$, where a, b, c are the lengths of the sides of a triangle, then the triangle is

A. right-angled
B. obtuse angled
C. equilateral
D. isosceles

Question No. 23

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

On which of the following lines lies the point of intersection of the line, $\frac{x-4}{2} = \frac{y-5}{2} = \frac{z-3}{1}$ and the plane, $x + y + z = 2$?

A. $\frac{x-4}{1} = \frac{y-5}{1} = \frac{z-5}{-1}$
B. $\frac{x-1}{1} = \frac{y-3}{2} = \frac{z+4}{-5}$
C. $\frac{x-2}{2} = \frac{y-3}{2} = \frac{z+3}{3}$
D. $\frac{x+3}{3} = \frac{4-y}{3} = \frac{z+1}{-2}$

Question No. 24

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If f be a continuous function on [0, 1], differentiable in (0, 1) such that f(1) = 0, then there exists some c $\in$(0, 1) such that

A. $c\, f'(c) - f(c) = 0$
B. $f'(c) + cf(c) = 0$
C. $f'(c) - cf(c) = 0$
D. $c\, f'(c) + f(c) = 0$

Question No. 25

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the integral $\int \frac{bx\cos 4x - a\sin 4x}{x^2}\, dx = \frac{a\sin 4x}{x} + c$, then the values of a and b are:

A. $a = 1, b = 4$
B. $a = -1, b = 4$
C. $a = 1, b = 1/4$
D. $a = 1/4, b = 2$

Question No. 26

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of ways in which 6 men and 5 women can dine at a round table if no two women are to sit together is given by

A. 30
B. $5! \times 4!$
C. $7! \times 5!$
D. $6! \times 5!$

Question No. 27

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \ldots + \binom{30}{20}\binom{30}{30}$ is equal to

A. ${}^{30}C_{11}$
B. ${}^{60}C_{10}$
C. ${}^{30}C_{10}$
D. ${}^{65}C_{55}$

Question No. 28

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The next term of the sequence $1,\ 3,\ 6,\ 10, \ldots$ is

A. 16
B. 13
C. 15
D. 14

Question No. 29

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $I_n = \int \tan^n x dx$ for $n \geq 2$, then $I_n + I_{n-2}$ is equal to

A. $\tan^n x + c$

B. $\frac{(\tan^{n-1} x)}{n-1} + c$

C. $\frac{\tan^n x}{n} + c$

D. $n\tan^n x + c$

Question No. 30

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area of the region (in sq. units), enclosed by the circle $x^2 + y^2 = 2$ which is not common to the region bounded by the parabola $y^2 = x$ and the straight line $y = x$, is

A. $\frac{1}{6}(24\pi - 1)$

B. $\frac{1}{3}(6\pi - 1)$

C. $\frac{1}{3}(12\pi - 1)$

D. $\frac{1}{6}(12\pi - 1)$

Question No. 31

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let y = y(x) be solution of the differential equation $\log_e(\frac{dy}{dx}) = 3x + 4y$, with y(0) = 0 . If $y(-\frac{2}{3}\log_e 2) = \alpha \log_e 2$ then the value of α is equal to:

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. 2

D. $-\frac{1}{2}$

Question No. 32

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the volume of a parallelopiped, whose coterminous edges are given by the vectors $\vec{a} = \hat{i} + \hat{j} + n\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} - n\hat{k}$ and, $\vec{c} = \hat{i} + n\hat{j} + 3\hat{k}$ $(n \geq 0)$ is 158 cubic units, then :

A. $\vec{a} \cdot \vec{c} = 17$
B. $\vec{b} \cdot \vec{c} = 10$
C. $n = 7$
D. $n = 9$

Question No. 33

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$$\lim_{n \to \infty} \left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + \ldots\ldots + \frac{(2n)^{1/3}}{n^{4/3}} \right)$$ is equal to

A. $\frac{3}{4}(2)^{4/3} - \frac{3}{4}$
B. $\frac{4}{3}(2)^{3/4}$
C. $\frac{4}{3}(2)^{4/3}$
D. $\frac{3}{4}(2)^{4/3} - \frac{4}{3}$

Question No. 34

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a certain population 10% of the people are rich, 5% are famous and 3% are rich and famous. The probability that a person picked at random from the population is either famous or rich but not both, is equal to

A. 0.07
B. 0.08
C. 0.09
D. 0.12

Question No. 35

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Angle between the tangents to the curve $y = x^2 - 5x + 6$ at the points (2, 0) and (3, 0) is

A. $\pi/2$
B. $\pi/3$
C. $\pi/6$

D. $\pi/4$

Question No. 36

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

There are $(p+q)$ different books on different topics in Mathematics, where $p \neq q$. If $L =$ The number of ways in which these books are distributed between two students X and Y such that X get p books and Y gets q books. $M =$ The number of ways in which these books are distributed between two students X and Y such that one of them gets p books and another gets q books. $N =$ The number of ways in which these books are divided into two groups of p books and q books, then

A. $L = M = N$
B. $L = 2M = 2N$
C. $2L = M = 2N$
D. $L = M = 2N$

Question No. 37

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let A be a square matrix of order 3 such that transpose of inverse of A is A itself, then |adj (adj A)| is equal to

A. 9
B. 27
C. 4
D. 1

Question No. 38

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The equations $2x + 3y = 7, 14x + 21y = 49$ has

A. infinitely many solutions
B. a unique solution
C. no solution
D. finitely many solutions

Question No. 39

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a variable chord PQ of the parabola $y^2 = 4ax$ subtends a right angle at the vertex, then the locus of the point of intersection of the normal at P and Q is

A. a parabola.
B. a hyperbola.
C. a circle.
D. an ellipse.

Question No. 40

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The function $f(x) = |2\text{sgn}(2x)| + 2$ is (where $\text{sgn}(x)$ means signum x)

A. differentiable at $x = 0$
B. discontinuous at $x = 0$
C. continuous at $x = 0$
D. none of these

English and LR (Maximum Marks: 90)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the given question, the given statement is "An annually conducted nationwide survey by a leading health research organization shows a continuing marked decline in the use of illegal drugs like hashish and charas by high school seniors over the last five years."

Which of the following, if true, would provide most support for concluding from the survey results described above that the use of illegal drugs by people below the age of 20 is declining?

A. Another survey found an increase in the rate of smoking amongst the high school seniors.

B. In the past, high school seniors were consistently the population group most likely to use illegal drugs and most likely to use them heavily.

C. The percentage of high school seniors who use illegal drugs is consistently very similar to the percentage of all people below the age of 20 who use illegal drugs.

D. Both (2) and (3).

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Most citizens are very conscientious about observing a law when they can see the reason behind it. For instance, there has been very little need to actively enforce the recently implemented law that increased the penalty for godmen duping people of their money by playing with their emotions. This is because citizens are very conscientious about duping someone in the name of religion, as it leaves their religious gurus with a bad name.

Which of the following statements would the author of this passage be most likely to believe?

A. The increased penalty alone is a significant motivation for most citizens to obey the law.

B. There are still too many inconsiderate citizens in the society.

C. Godmen should not be allowed to play with the emotions of the people.

D. Society should make an effort to teach citizens the reasons for its laws

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Directions: In the question given below, a sentence with a blank is given. Choose the alternative that completes the sentence grammatically and contextually.

As you sow _____ shall you reap.

A. when

B. as

C. like

D. so

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Directions: In the question given below, a sentence with a blank is given. Choose the alternative that completes the sentence grammatically and contextually.

_____ lion is _____ king of beasts.

A. a
B. an
C. the, an
D. a, the

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find out the one word among the options which can not be formed by using the letters of the word given in the question.

DEVASTATING

A. DATING
B. VAST
C. STING
D. TEARING

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Choose the word which is nearly opposite in meaning to the given word: 'CALUMNY'

A. Apology
B. Eulogy
C. Enjoyment
D. Reservation

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the following question, a word is followed by four options. You have to choose the option that is similar in meaning to the given word.

Doleful (feeling sorrow)

A. Bright
B. Jovial
C. Morbid
D. Mournful

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the following question, a word is followed by four options. You have to choose the option that is similar in meaning to the given word.

Rescind (to repeal)

A. Cancel
B. Enjoy
C. Praise
D. Receive

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Fill in the blanks with an appropriate determiner.

She prefers small get-togethers as she does not have _____ friends.

A. Anyone
B. Many
C. Much
D. Some

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Every sentence (given below) is divided into three sections marked A, B, and C. Read the sentence carefully and mark the section that contains an error. If there is no error in the sentence, mark (D).
I declared (A)/that all will go wrong (B)/without her. (C)/no error. (D)

A. A
B. B
C. C
D. D

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Select the related letters/words/numbers from the given alternatives.

XXIV : 48 :: XIV?

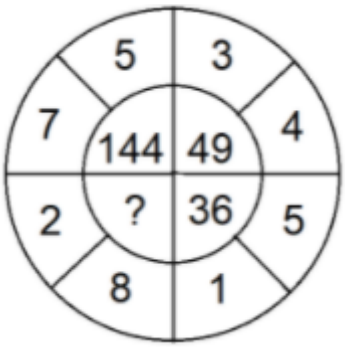
A. 28
B. 24
C. 14
D. 30

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the missing number.



A. 82
B. 124
C. 68
D. 100

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

52, 45, 66, 31, 80, ?

A. 17
B. 16
C. 15
D. None of these

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Identify the next number in the series 2, 7, 17, 32, 52, 77, ___.

A. 107
B. 98
C. 101
D. 92

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Directions: Find the odd one out.

A. Goose
B. Crocodile
C. Toad
D. Chicken

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

DIRECTIONS: In each of the following questions, four alternatives are given, out of which three are alike in a certain way while one is different. Choose the odd one.

A. 9611
B. 7324
C. 2690
D. 1754

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Pointing towards A, B said, he is the only brother of my father's wife. Who is A to B?

A. Maternal uncle
B. father
C. Paternal Uncle
D. Nephew

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Pointing to Venkat, Vasantha said, "He is the son of my father's son". How is Venkat's mother related to Vasantha?

A. Daughter
B. Sister-in-law
C. Sister
D. Aunt

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

DIRECTIONS: In each of the following questions, four alternatives are given, out of which three are alike in a certain way while one is different. Choose the odd one.

A. Jupiter
B. Saturn
C. Venus
D. Sun

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following pair is not correct?

A. 11 − 1320
B. 15 − 3360
C. 14 − 2730
D. 18 − 5812

Question No. 21

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Sumit is facing east. He hears a sound and turns 135° in the anti-clockwise directions. Then he turns 90° in the clockwise direction. Which direction is he facing now?

A. North
B. East
C. North-east
D. South-east

Question No. 22

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Directions: Read the given information carefully and answer the questions given below:
Megha walks 10 km towards North. She turns right and walks 15 km. She turns right and walks 20 km. She turns right and walks 15 km. How far (in km) is she from her starting point?

A. 15
B. 5
C. 10
D. 20

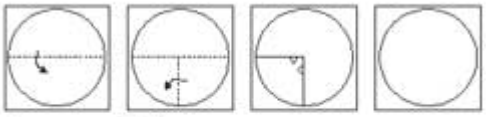
Question No. 23

Only One Option Correct Type

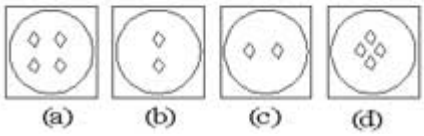
Each question has multiple options out of which ONLY ONE is correct.

Directions: In the following question, a piece of paper is folded and cut as shown below in the question figures. From the given answer figures, indicate how it will appear when opened.

Question Figures:



Answer Figures:



A. A
B. B
C. C
D. D

Question No. 24

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A word is represented by only one set of numbers as given in any one of the alternatives. The sets of numbers given in the alternatives are represented by two classes of alphabets as in two Matrixes given below. The columns and rows of Matrix I are numbered from 1 to 5 and that of Matrix II are numbered from 6 to 10. A letter from these matrices can be represented first by its row and next by its column e.g., 'A' can be represented by 11, 23 etc. and 'G' can be represented by 67, 78, 75 etc. Similarly, you have to identify the set for the word given below:

BEE

Matrix I

	1	2	3	4	5
1	A	B	C	D	E
2	E	D	A	B	C
3	B	C	D	E	A
4	D	A	E	C	D
5	C	E	B	A	B

Matrix II

	6	7	8	9	10
6	F	G	H	I	J
7	J	I	G	H	F
8	F	H	I	J	G
9	G	J	F	G	I
10	H	E	J	F	E

Matrix I

	1	2	3	4	5
1	A	B	C	D	E
2	E	D	A	B	C
3	B	C	D	E	A
4	D	A	E	C	D
5	C	E	B	A	B

Matrix II

	6	7	8	9	10
6	F	G	H	I	J
7	J	I	G	H	F
8	F	H	I	J	G
9	G	J	F	G	I
10	H	E	J	F	E

A. 12, 15, 41
B. 12, 21 15
C. 12, 15, 33
D. 21, 12, 22

Question No. 25

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If in the certain word ‘DICKINSON’ is written as ‘357950210’ and ‘DIPP’ is written as ‘3566’, then write ‘PICNIC’ using the same code?

A. 657507
B. 560757
C. 650757
D. 657057

Question No. 26

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A series is given with one term missing. Select the correct alternative from the given ones that will complete the series.

sam, tbn, uco,?

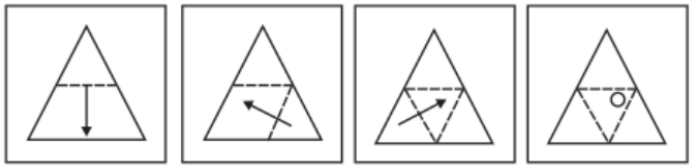
A. vwx
B. sap
C. vdp
D. vde

Question No. 27

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A triangular sheet of paper has been folded and punched as shown below. How will it appear when opened?



A.

B.

C.

D.

Question No. 28

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the following question, various terms of an alphabet series are given with one or more terms missing as shown by (?). Choose the missing terms out of the given alternatives.

M, N, O, L, R, I, V,?

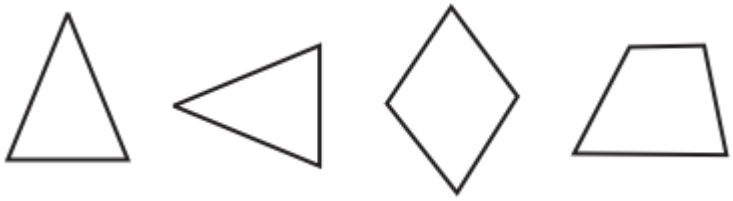
A. A
B. E
C. F
D. H

Question No. 29

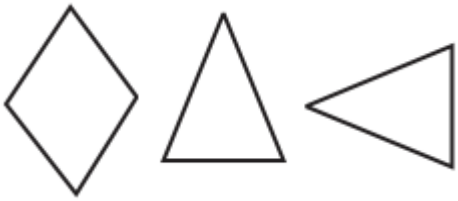
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

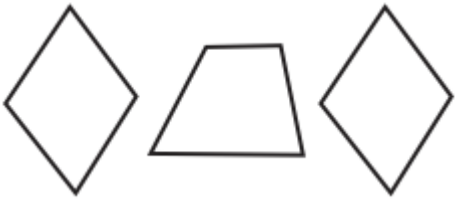
Select that combination of the given parts which can fit in to each other to form an equilateral triangle.



A.

B. 

C.

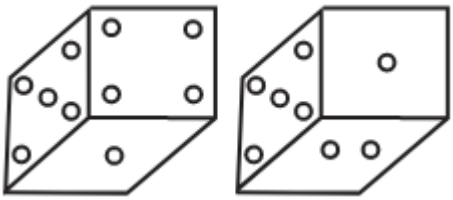
D. 

Question No. 30

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Six surfaces of a dice are shown below. Which will be the opposite of four dots pointed face?



A. 2

B. 4
C. 5
D. 3